

Digital Systems

Tutorial 2

25 March 2023

Tutorial Questions(1-13)

1. Simplify the Boolean expressions using 3 variable K-maps.

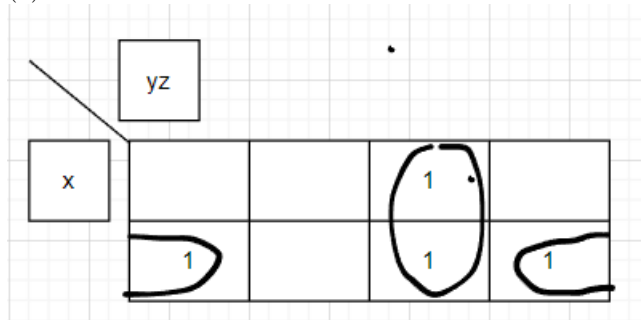
(a) $F(x, y, z) = \sum(3, 4, 6, 7)$

(b) $F(x, y, z) = \sum(0, 2, 4, 5, 6, 7)$

(c) $F(x, y, z) = \sum(1, 3, 5, 7, 6)$

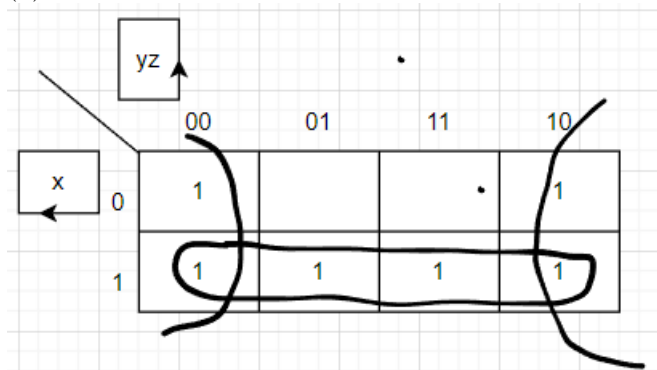
Ans. The following figures represent K-map and the simplified min terms expressions.

(a)



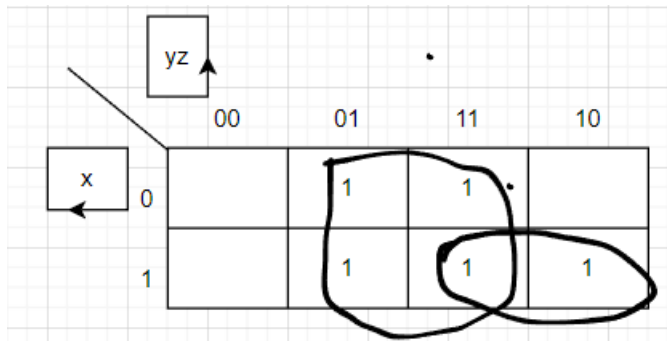
The two groups give the minimized expression as $F = yz + xz'$

(b)



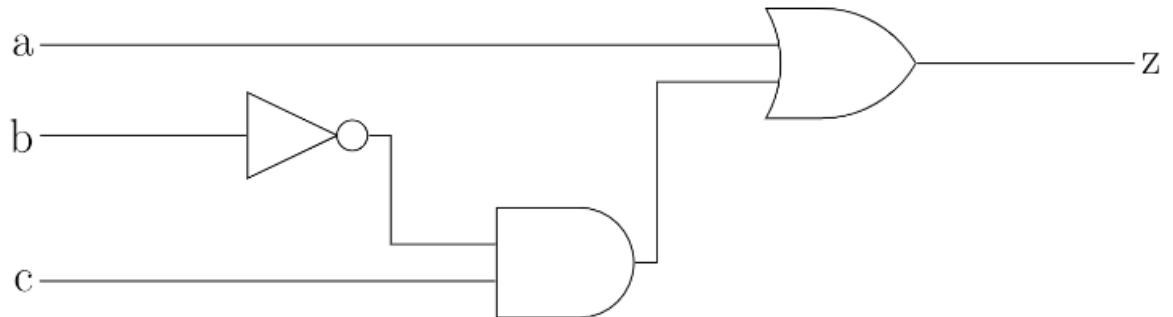
In this K-map, we can group two quads, and the minimized expression is $F = x + z'$

(c)



In this K-map, we can group 1 quad and one pair as shown in figure, and the minimized expression is $F = z + xy$

2. consider the boolean function $z(a, b, c)$. Which of the following minterms represent this circuit.



- (a) $z = \sum(0, 1, 3, 7)$
 (b) $z = \sum(1, 4, 5, 6, 7)$
 (c) $z = \sum(2, 4, 5, 6, 7)$
 (d) $z = \sum(2, 3, 5)$

Answer

From given circuit $z = a + b'c$

K-Map for the above expression is:

		bc			
		00	01	11	10
a	0	0	1	0	0
	1	1	1	1	1

Minterms are $\sum(1, 4, 5, 6, 7)$

Hence, option (B) is correct

3. Using a 4-variable K-map find the minimal SOP expressions for the following.

(a) $F(a, b, c, d) = \sum(3, 5, 7, 8, 10, 11)$

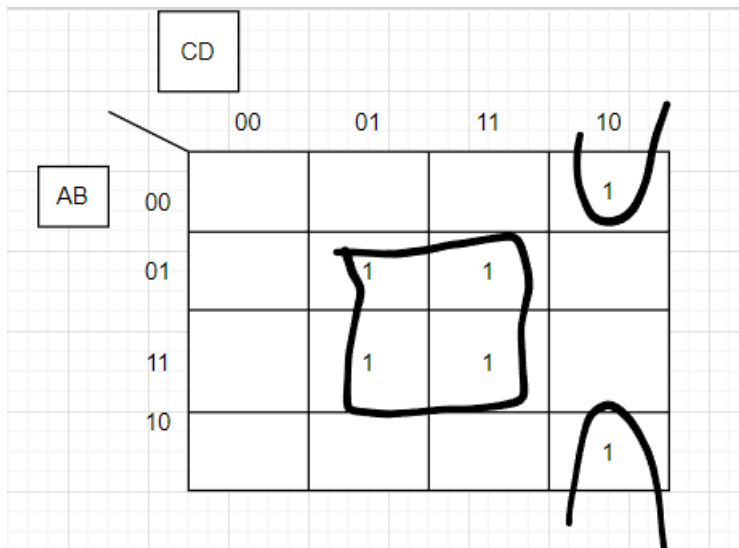
(b) $F(a, b, c, d) = \sum(5, 7, 13, 15, 2, 10)$

Ans. (a)

		CD			
		00	01	11	10
AB	00			1	
	01		1	1	
	11				
	10	1		1	1

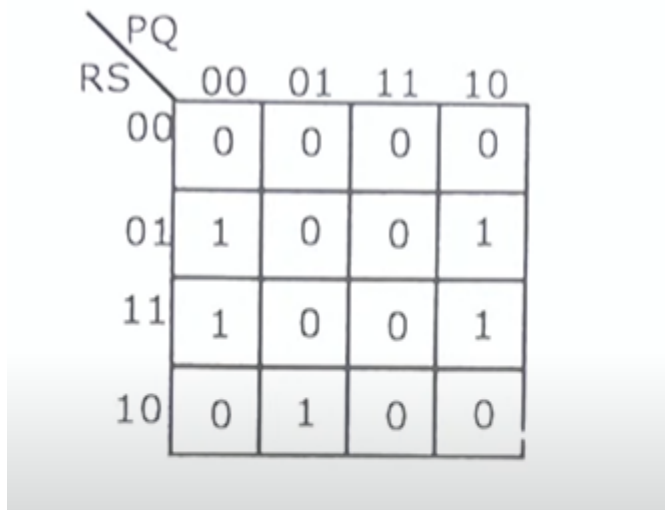
The minimized SOP output expression is $F = ab'd' + a'bd + b'cd$

(b)



The minimized POS output expression is $F = b'cd' + bd$

4. i) The K-map for a four variable Boolean function is given in figure. The sum of product expression?



(ANS= $Q'S + P'QRS'$)

Q The k-map for a four variable Boolean function is given in figure. The expression for the sum of product expression is?

A)

		00	01	11	10
RS	00	0	0	0	0
	01	1	0	0	1
	11	1	0	0	1
	10	0	1	0	0

$F = \bar{Q}S + \bar{P}QRS$

ii) The minimum sum of products form of the Boolean expression using K-map is—?
 $Y = P'Q'R'S' + PQ'R'S' + PQ'R'S + PQ'RS + PQ'RS' + P'Q'RS'$ (ANS= $Q'S' + PQ'$)

Q The minimum sum of products form of the Boolean expression using k-map is — ?

$$Y = \bar{P}\bar{\theta}\bar{R}\bar{S} + P\bar{\theta}\bar{R}\bar{S} + P\bar{\theta}\bar{R}S + P\bar{\theta}RS + P\bar{\theta}R\bar{S} + \bar{P}\bar{\theta}R\bar{S}$$

A)

	RS			
P $\bar{\theta}$	00	01	11	10
00	1			1
01				
11				
10	1	1	1	1

$$Y = \bar{\theta}\bar{S} + P\bar{\theta}$$

5. For $f(a, b, c, d) = \pi(0, 1, 3, 7)$

The minimum number of terms in minimal SOP form is assumed to be 'X'. The value of 'X' must be...? ANS= 3

Q. For $f(A,B,C,D) = \pi(0,1,3,7)$.

The minimum number of terms in minimal SOP form is assumed to be "X". The value of "X" must be —?

A)

AB \ CD	00	01	11	10
00	0	0	0	1
01	1	1	0	1
11	1	1	1	1
10	1	1	1	1

$$f = A + B\bar{C} + C\bar{D}$$

$$\& \quad X = 3$$

6. Simplify the following functions using k-map.

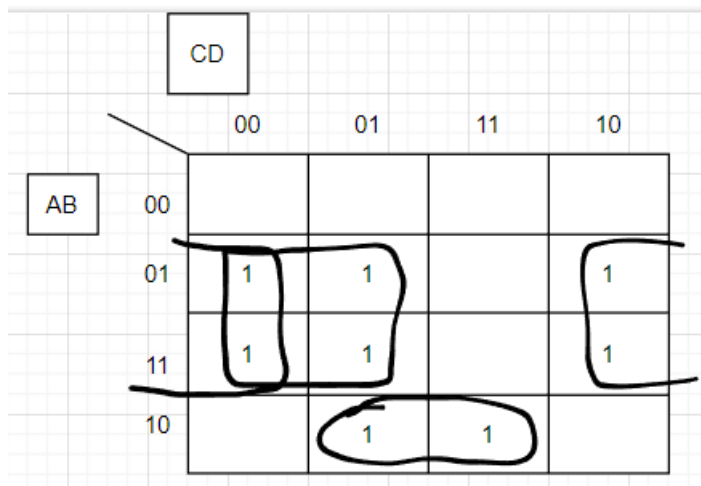
(a) $BD' + AB'CD + AC'D + BC'$

(b) $A'B'C'D + CD' + AC'D$

ans. (a) To solve the expressions, we need to find the truth table and then using the truth table we can implement our K-map to find the minimized expression.



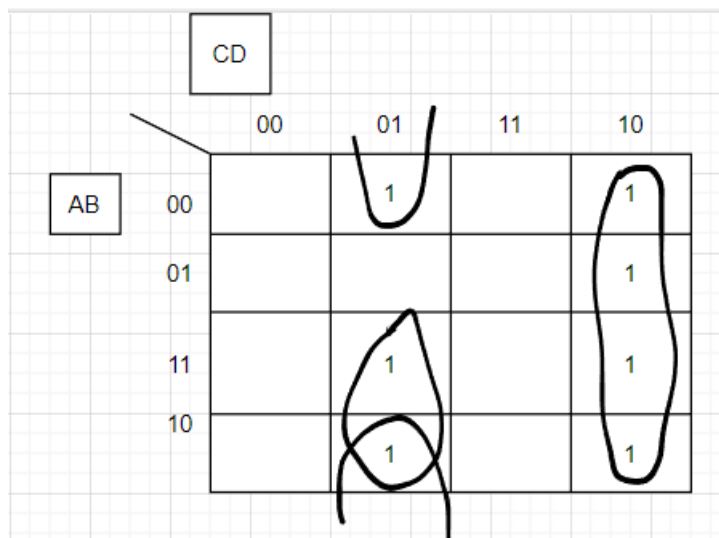
A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



The minimized expression obtained from the K-map $BC' + BD' + AB'D$
(b)



A	B	C	D	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



The minimized expression obtained from k-map is $CD' + B'C'D + AC'D$

7. Simplify the Boolean expressions .

(a) $A(A+B'C'+C)+B'(C+A'+BC)(A+B'C+AC')=1$, if $C=A'$ then $A+B'=?$

(b) If $Y= A \text{ XOR } B \text{ XOR } AB$, then the simplified basic gate expression is $Y = A ? B$

① a) In the logic equation given below

$$A(A + \bar{B}\bar{C} + C) + \bar{B}(C + \bar{A} + BC)(A + \bar{B}C + A\bar{C}) = 1,$$

if $C = \bar{A}$ then $A + \bar{B} = \underline{\hspace{1cm}} ?$

A) Given $A(A + \bar{B}\bar{C} + C) + \bar{B}(C + \bar{A} + BC)(A + \bar{B}C + A\bar{C}) = 1$

Replacing $C = \bar{A}$

$$\Rightarrow A(A + \bar{B}\bar{A} + \bar{A}) + \bar{B}(\bar{A} + \bar{A} + B\bar{A}) + (A + \bar{B}\bar{A} + A\bar{A}) = 1$$

$$A(1 + \bar{A}\bar{B}) + \bar{B}(\bar{A}(1 + B)) + (A + \bar{A}\bar{B}) = 1$$

$$A + \bar{B}\bar{A} + A + \bar{B} = 1$$

$$\boxed{A + \bar{B} = 1}$$

Note: we know $A + \bar{A}B = A + B$

$$A + \bar{B} = A + B$$

② b) If $Y = A \oplus B \oplus AB$, then the simplified basic gate expression is $Y = A \underline{\hspace{1cm}} ? \underline{\hspace{1cm}} B$

A) Truth table of 'Y'

A	B	AB	Y
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

OR gate
 $Y = A + B$
 ~~$Y = A \oplus B$~~

8. For the Boolean expression given below, what is the minimized product of sum (POS) expression using K-map?

$$f = a'b'c' + a'bc' + ab'c' + abc + abc'$$

Q For the Boolean expression given below
 $f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$, the minimized
 product of sum (POS) expression using K-map is - ?

A)

	bc	00	01	11	10
a	0	1	0	0	1
	1	1	0	1	1

$f = (b + \bar{c})(a + \bar{c})$

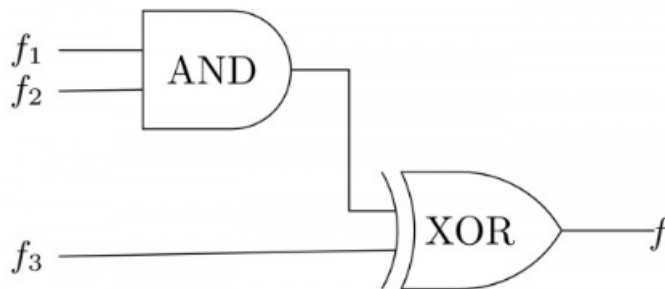
9 Consider three 4 variable functions given below.

$$F_1 = \sum(0, 2, 5, 8, 14)$$

$$F_2 = \sum(2, 3, 6, 8, 14, 15)$$

$$F_3 = \sum(2, 7, 11, 14)$$

for the following circuit with one And gate and one XOR gate find out the output expression(F)?



Answer

f_1 and $f_2 = \sum(0,2,5,8,14)$ and $\sum(2,3,6,8,14,15) = \sum(2,8,14)$ (common min terms)

$(f_1 \text{ and } f_2) \oplus f_3 = \sum(2,8,14) \oplus \sum(2,7,11,14)$

we know $a \oplus b = a'b + ab'$, so

$\sum(2,8,14) \oplus \sum(2,7,11,14)$

$= (\sum(0,1,3,4,5,6,7,9,10,11,12,13,15) \text{ and } \sum(2,7,11,14))$ or $(\sum(2,8,14) \text{ and } \sum(0,1,3,4,5,6,8,9,10,12,13,15))$

$= \sum(7,11) \text{ or } \sum(8)$

$= \sum(7,11) \text{ or } \sum(8)$

$= \sum(7,8,11)$ (union)

10. (1) Find the minimized sop expression for the given K-maps.

(2) find the number of prime implicants and essential prime implicants and draw the them in separate K-maps.

(a)

		CD			
		00	01	11	10
AB	00				
	01	1	1	1	1
	11		1	1	1
	10		1		

(b)

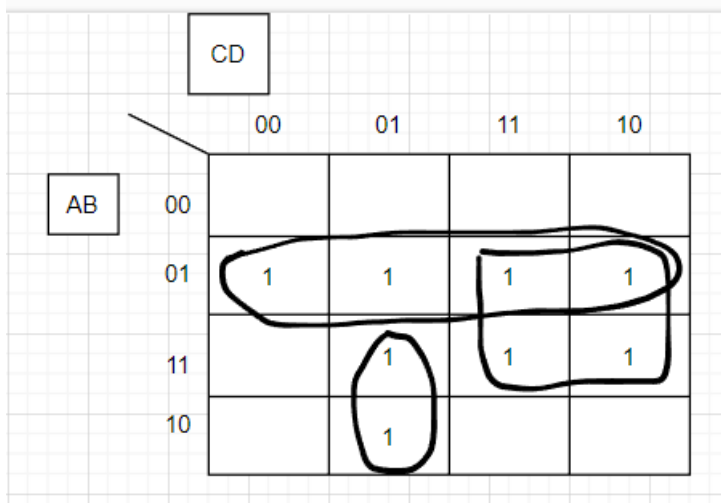
		CD			
		00	01	11	10
AB	00		1		
	01		1	1	1
	11	1	1	1	
	10			1	

ans.

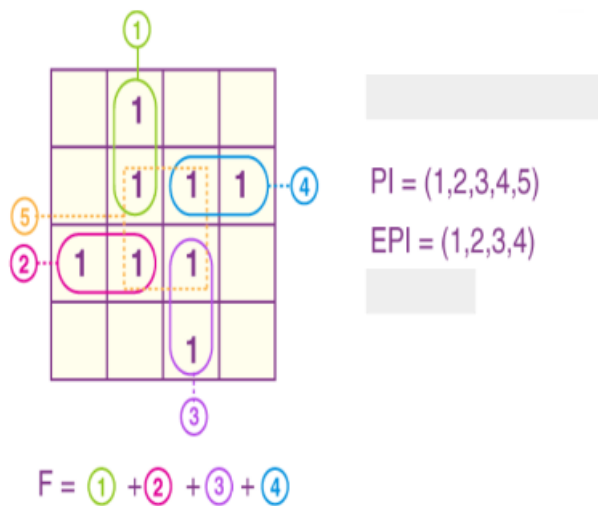
(a)

		CD			
		00	01	11	10
AB	00				
	01	1	1	1	1
	11		1	1	1
	10		1		

The prime implicants is shown in the above figure here. The middle quad is redutant and total number of prime implicants are 4.



The minimized expression in this case is $f(A, B, C, D) = A'B + BC + AC'D$. The number of EPI is 3.



The minimized expression is $f(A, B, C, D) = A'C'D + A'BC + ACD + ABC'$

- 11 find out essential prime implicants and prime implicants of the following Boolean function?
 $F(a, b, c) = a'c + ac' + b'c$

Answer

$$f(a, b, c) = a'c + ac' + b'c$$

We can write these product of sum terms into canonical product of sum form.

$$f(a, b, c) = \underbrace{a'bc}_{011} + \underbrace{ab'c'}_{100} + \underbrace{abc'}_{110} + \underbrace{ab'c}_{101} + \underbrace{a'b'c}_{001}$$

$$f(a, b, c) = \sum(1, 3, 4, 5, 6)$$

Now, we can draw the k-map for these minterms.

		bc			
		00	01	11	10
a	0	0	1	1	0
	1	1	1	0	1

- Prime implicant of f is an implicant that is minimal - that is, the removal of any literal from product term results in a non-implicant for f .
- Essential prime implicant is an prime implicant that cover an output of the function that no combination of other prime implicants is able to cover.

Prime implicants are: $a'c, b'c, ab', ac'$

Essential prime implicants are: $a'c, ac'$ (green color).

12. In the K-map shown below X denotes a don't care term. What is the minimal form of the function represented by the map?

		AB			
		00	01	11	10
CD	00	1	1		1
	01	X			
	11	X			
	10	1	1		X

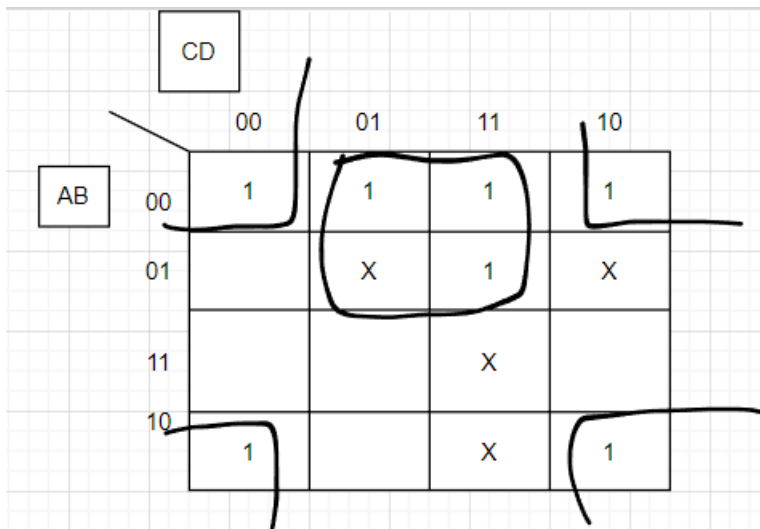
		AB			
		00	01	11	10
CD	00	1	1		1
	01	X			
	11	X			
	10	1	1		X

Ans.

We know that two do not care conditions can not be grouped together. So we are getting two quads as shown in the figure the minimized expression is $A'D' + B'D'$

- 13 Given $f(a, b, c, d) = \sum(0, 1, 2, 3, 7, 8, 10) + \sum_d(5, 6, 11, 15)$. Find the minimum SOP form for the expression. (The term $\sum_d$ represents the do not care condition)

answer

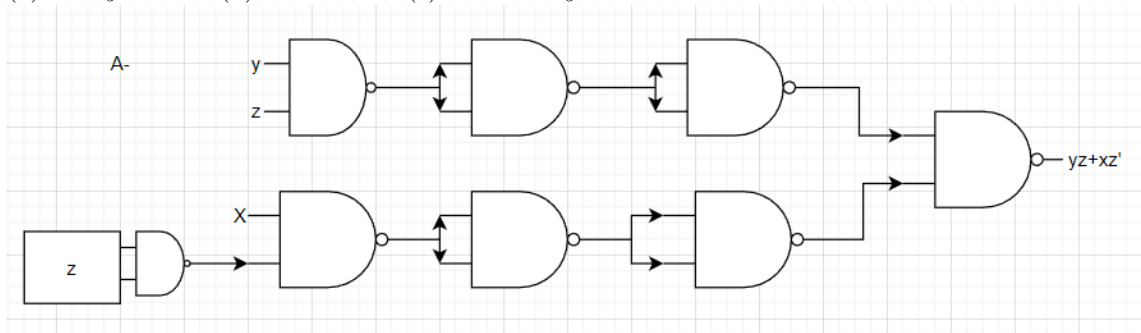


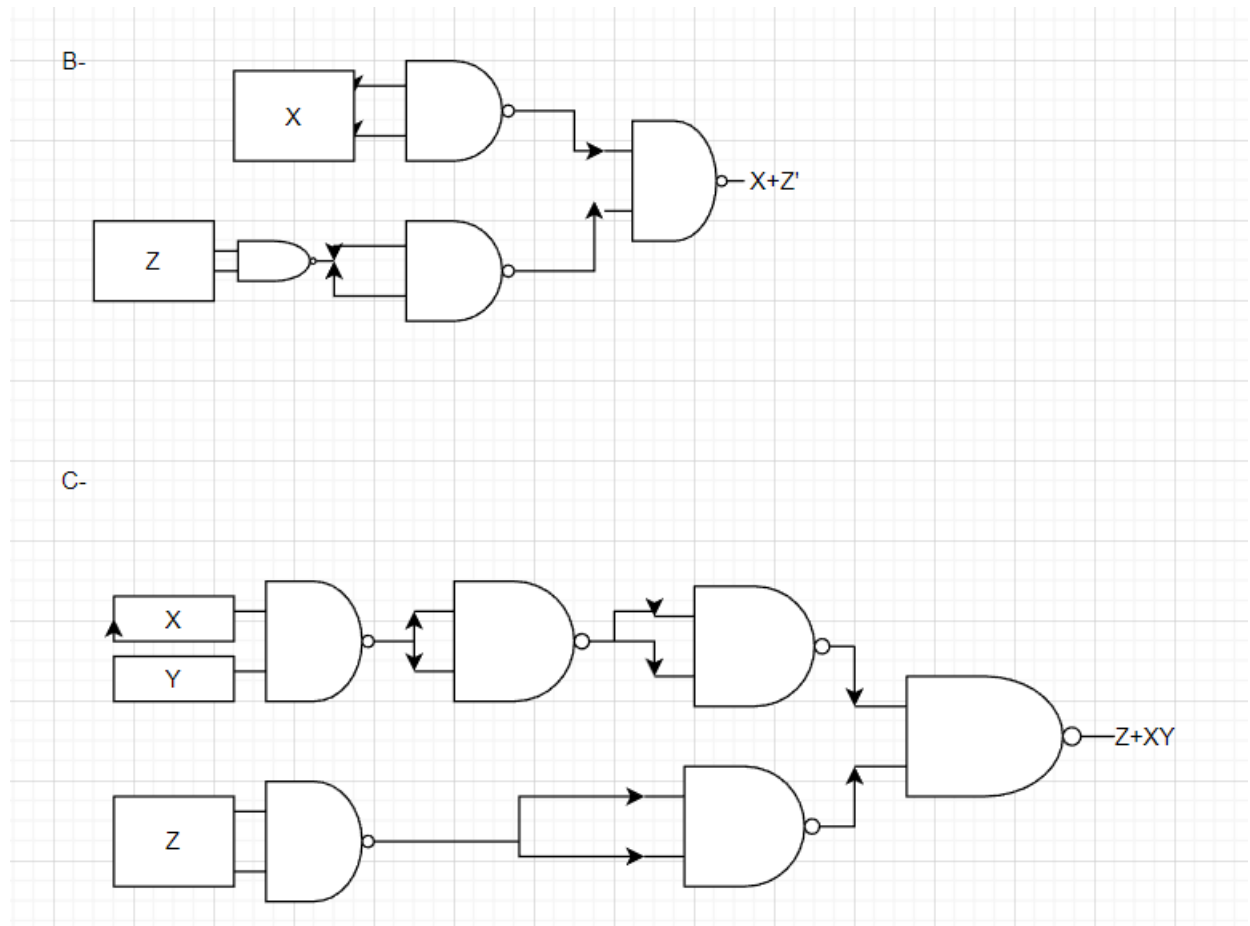
Here we formed two quads using do care condition and oness. The minimized expression for it is $f = B'D' + A'D$

- **Practice Questions(14-20)**

14. Implement the solutions of question number-1 using 2 input NAND gates.

ans (a) $F = yz + xz'$ (b) $F = x + z'$ (c) $F = z + xy$

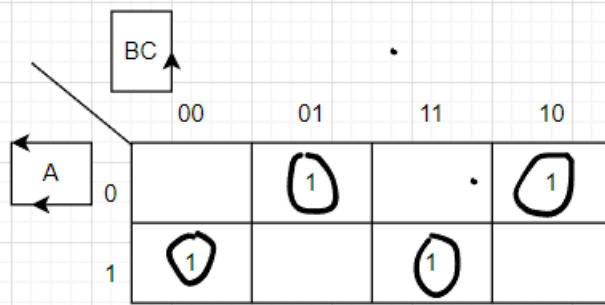




15 (a) Solve the following truth table using K-maps. (b) Implement it using XOR gates

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Ans



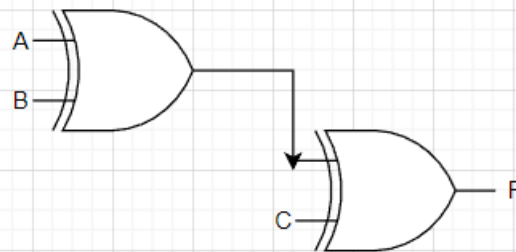
No groups can be made here.

$$P = \bar{A} \bar{B} C + \bar{A} B \bar{C} + A \bar{B} \bar{C} + A B C$$

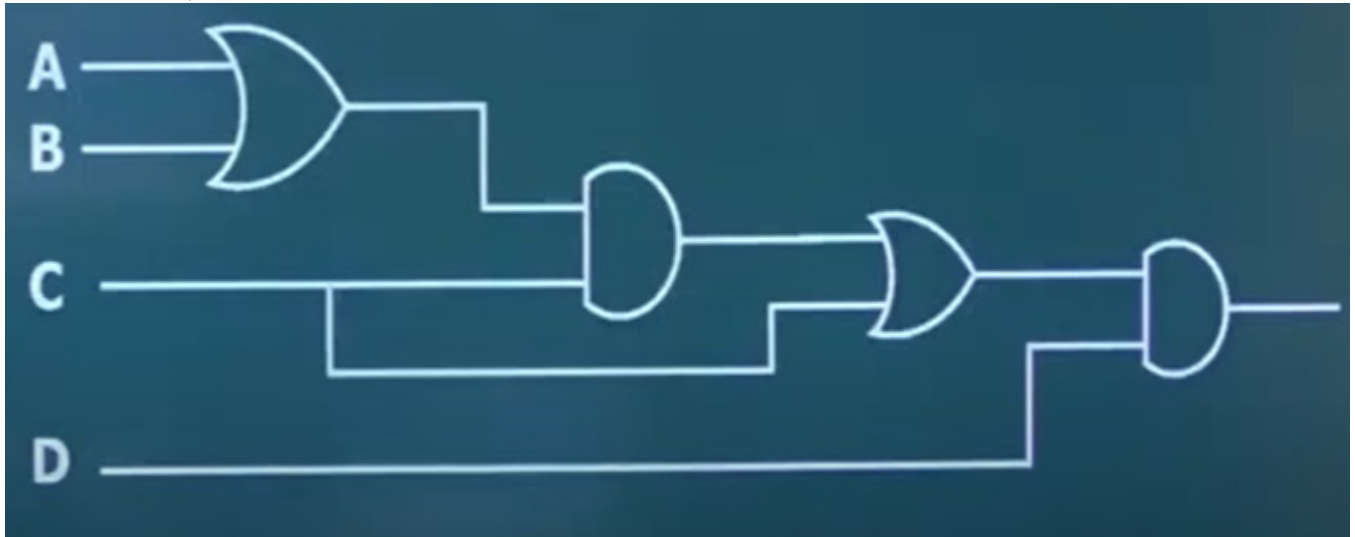
$$= \bar{A} (\bar{B} C + B \bar{C}) + A (\bar{B} \bar{C} + B C)$$

$$= \bar{A} (B \oplus C) + A (\overline{B \oplus C})$$

$$P = A \oplus B \oplus C$$

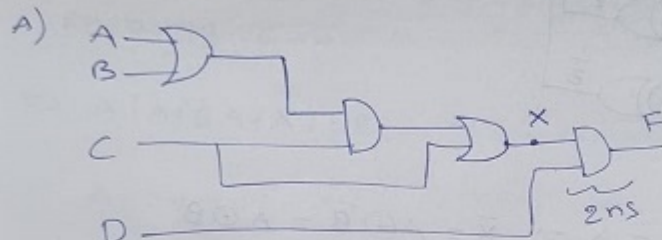


16. (a) What is the minimum and maximum propagation delay of circuit given below ? (Delay of AND OR gates is 2ns)



- (b) $F1 = \sum m(0, 4, 5, 6)$ $F2 = \sum m(0, 3, 6, 7)$. Find the minterm expansion for $F1'F2'$?
 (Note: Consider Don't care cases are not available) (Ans = $\sum m(1, 2)$)

- What is the minimum propagation delay of circuit given below and under what possible condition? (Delay of AND & OR gates is 2ns)



if $D=0 \Rightarrow F = X \cdot D$
 $= X \cdot 0 = 0$ (independent of X)

$\therefore$ minimum propagation delay = 2ns

& condition is $\boxed{D=0}$

- Given $F_1 = \sum m(0,4,5,6)$ & $F_2 = \sum m(0,3,6,7)$.

Find the minterm expansion for $\overline{F_1} \overline{F_2}$

Note: Consider we do not have any don't care cases.

A) $\overline{F_1} = \sum m(1,2,3,7)$ & $\overline{F_2} = \sum m(1,2,4,5)$

$\overline{F_1} \overline{F_2} \Rightarrow \overline{F_1} \cap \overline{F_2} = \sum m(1,2)$

17. The minimized function that can detect a divisible by '3', in 8421 BCD code digit is given by?
 (Represent in terms of $D_8 D_4 D_2 D_1$)

a.

The minimum function that can detect a divisible by 3!

8421 BCD code digit (representation is $D_8 D_4 D_2 D_1$) is given by

A)

D_8	D_4	D_2	D_1	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	X
⋮				⋮
1	1	1	1	X

$D_8 \backslash D_4$	$D_2 D_1$	00	01	11	10
00		1			
01					1
11		X	X	X	X
10			1	X	X

$$Y = \bar{D}_8 \bar{D}_4 \bar{D}_2 \bar{D}_1 + D_8 D_1 + D_4 D_2 \bar{D}_1 + \bar{D}_4 D_2 D_1$$

18. i) The K-map for a four variable Boolean function is given in figure. The sum of product expression?
'X'—Don't care notation

CD \ AB	AB			
	00	01	11	10
00		1	X	
01			X	
11	1	1	X	X
10	1		X	X

Q The minimized expression for the k-map of given below is

A)

CD \ AB	AB			
	00	01	11	10
00		1	X	
01			X	
11	1	1	X	X
10	1		X	X

X — don't care

A
$$F = CD + \bar{B}C + B\bar{C}\bar{D}$$

- ii) The K-map for a five variable (P, Q, R, S, X) Boolean function is given in figure. The sum of product expression?

RS \ PQ	PQ			
	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

$X = 0$

RS \ PQ	PQ			
	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	1	1	0

$X = 1$

Following is the K-map of a Boolean function of five variables P, Q, R, S and X. The minimum sum-of-products (SOP) expression for the function is ?

RS \ PQ	PQ			
	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

$X = 0$

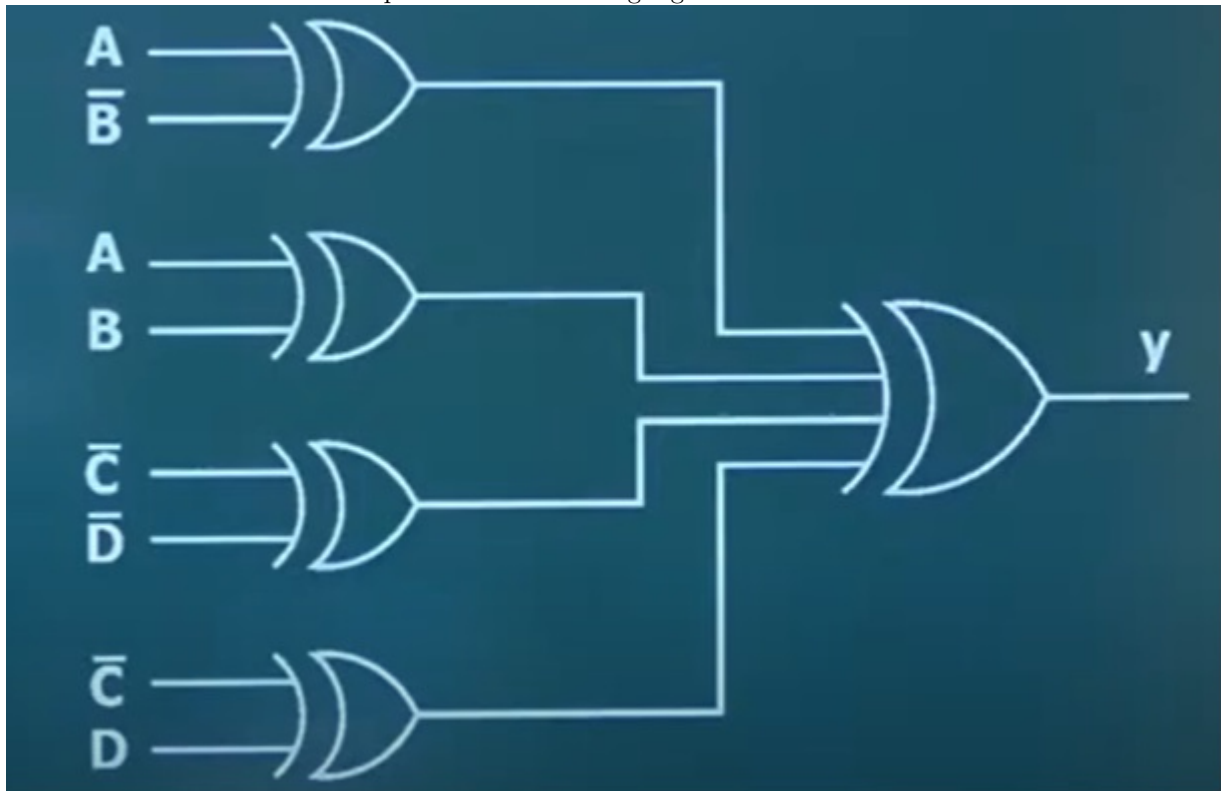
RS \ PQ	PQ			
	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	1	1	0

$X = 1$

A)

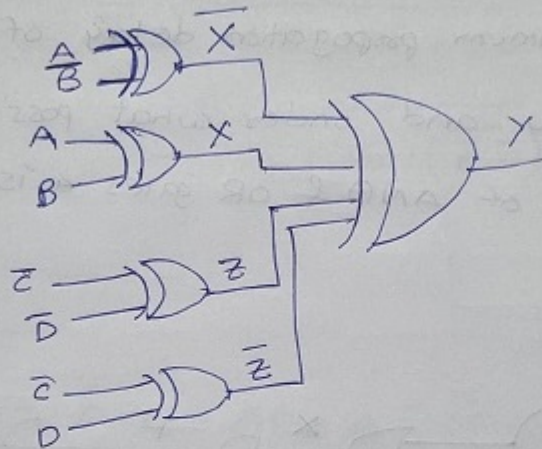
$$f = \bar{P}S\bar{X} + P\bar{S}X$$

19. What is the value of Y after simplification of following logic circuit?



a) what is the value of "Y" after simplification of following logic circuit?

1)



$$\text{let } A \oplus B = X \Rightarrow \bar{X} = A \oplus \bar{B} = A \odot B$$

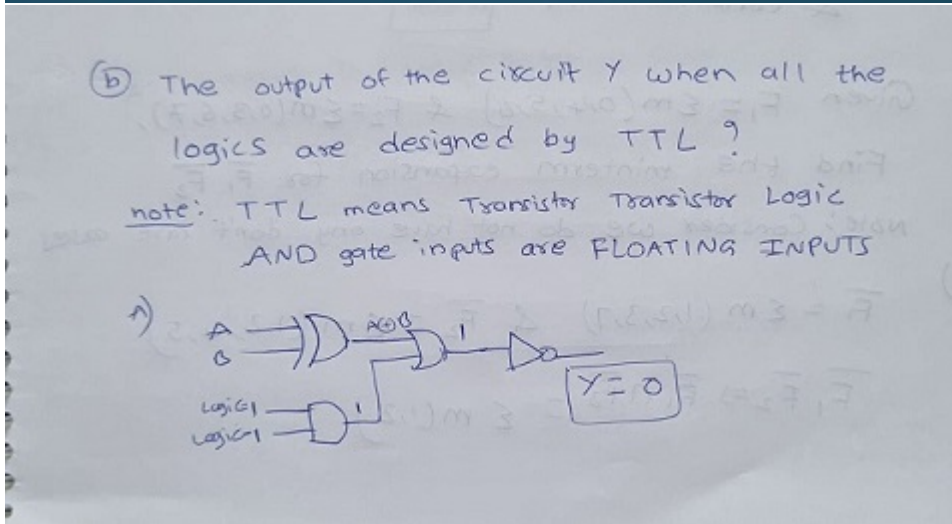
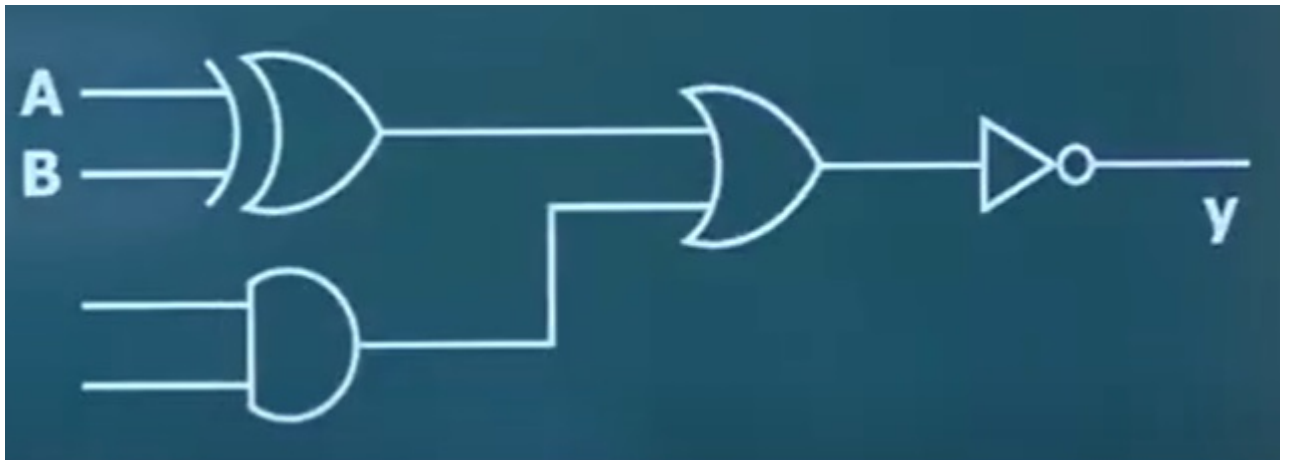
$$\text{let } \bar{C} \oplus \bar{D} = Z \Rightarrow \bar{Z} = \bar{C} \oplus D = \bar{C} \odot \bar{D}$$

$$Y = \bar{X} \oplus X \oplus Z \oplus \bar{Z}$$

$$Y = 0 \quad (\because \text{even number of same inputs})$$

20. The output of the circuit 'Y' when all the logic gates are designed by TTL?

Note: TTL means Transistor Transistor Logic AND gate has floating inputs



- Question No. (1,2,3,6,9,10,11,12,13,14,15)-Jagyansis
- Question No. (4,5,7,8,16,17,18,19,20)-Naveen